Water Security Plan Report

Date : 21/ 12/ 2023 Time : 5:48 PM User Name: Arkhi_155944

STATE: GUJARAT_24 » DISTRICT: BANASKANTHA_441 » BLOCK: DANTIWADA_3735 » GRAM PANCHAYAT: ARKHI_155944 » YEAR: 2023-24 »

Gram Panchayat Information

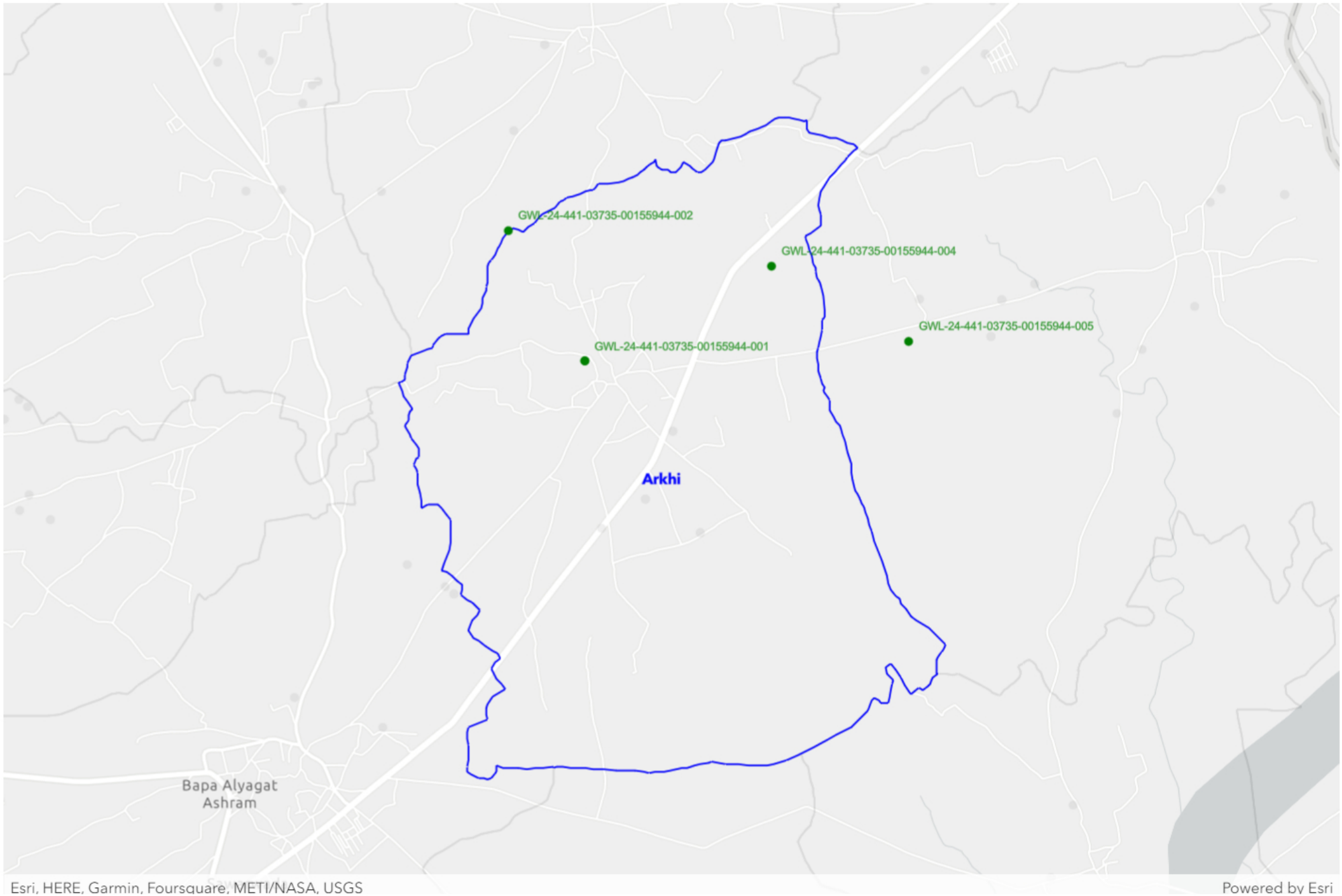
State:	Gujarat_24	District:	Banaskantha_441
Block / Taluk :	Dantiwada_3735	GP Name :	Arkhi_155944
GP ID :	155944	Block Area (Ha) :	45250.16
GP area (Ha) :	1045.77	Watershed Code :	SRWT003
Watershed name :	SARASWATI	Watershed area in Block / Taluk (Ha) :	131645
Sub-basin Code :	5B	Sub basin name :	Saraswati
Basin Code :	20	Basin Name :	Kutch

Location Map

Watershed Map

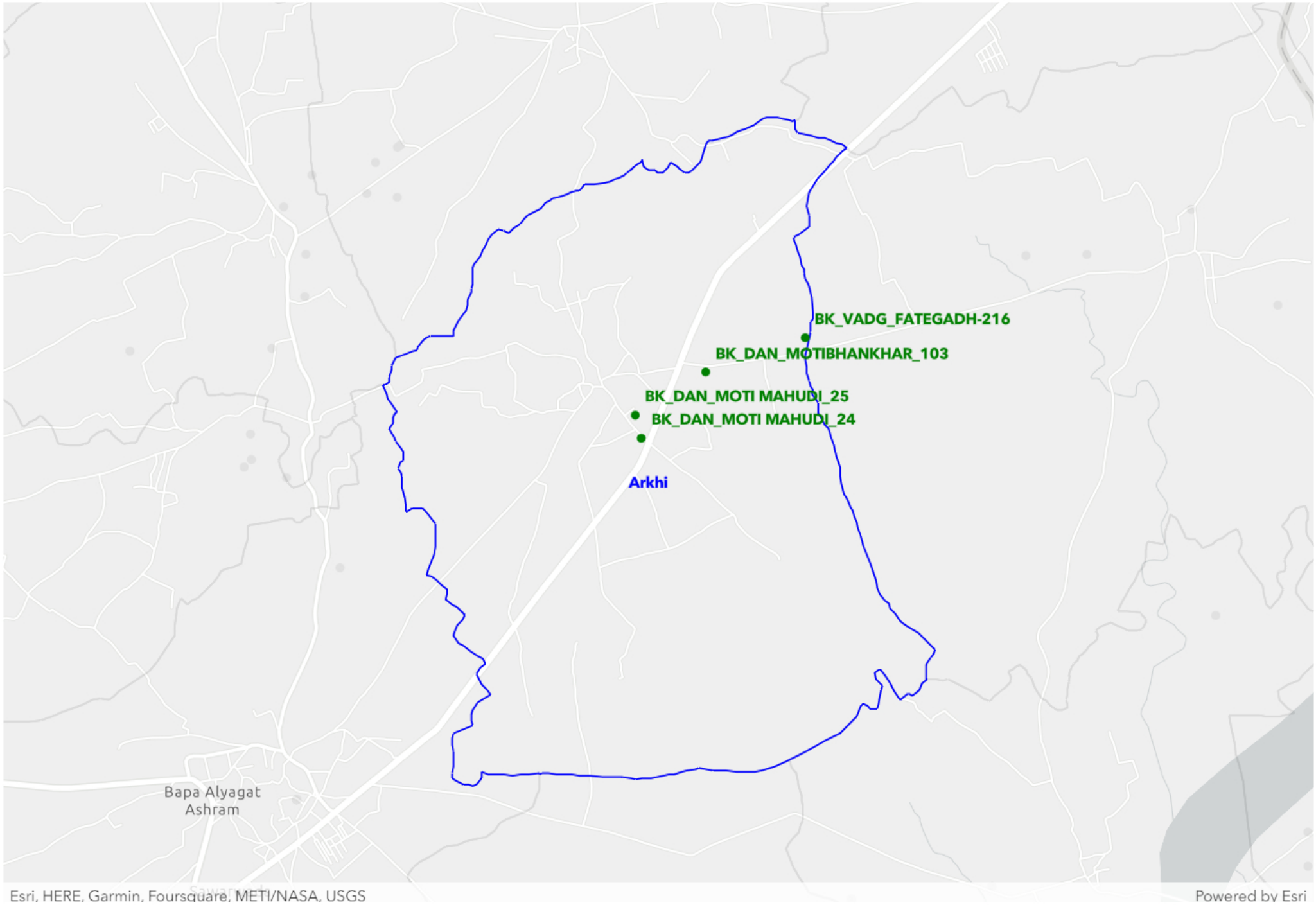
Aquifer Map

Ground Water Level



S.No.	Site Name	Location Details	Latitude	Longitude	Date	Well ID	Type of Well	Other Well Type	Ownership Type	Government Source	Diameter of well(m)	Depth of Well(mbmp)	Height of Measuring Point (mbgl)	Water Level (mbgl)	Water Level Date
1	Arkhi	Sujabhai chopabhai surala	24.51846	72.32056	2023-06-10 15:30:00	GWL-24-441-03735-00155944-001	Tube Well		Private		0.2000	160.00	1.00	50	2023-06-10
2	Arkhi	Jesungbhai dhudabhai surala	24.52644	72.31539	2023-06-10 15:40:00	GWL-24-441-03735-00155944-002	Tube Well		Private		0.20	125.00	1.00	65	2023-06-10
3	Arkhi	Haribhai jesungbhai surala	24.51846	72.32056	2023-06-10 15:45:00	GWL-24-441-03735-00155944-003	Tube Well		Private		0.2000	65.00	1.00	54	2023-06-10
4	Arkhi	Gajabhai chopabhai surala	24.52427	72.33315	2023-06-10 15:59:00	GWL-24-441-03735-00155944-004	Tube Well		Private		0.2000	85.00	1.00	40	2023-06-10
5	Arkhi	Arjanbhai ragabhai kuniya	24.51965	72.3424	2023-06-10 16:04:00	GWL-24-441-03735-00155944-005	Tube Well		Private		0.2000	65.00	1.00	50	2023-06-10

Ground Water Quality

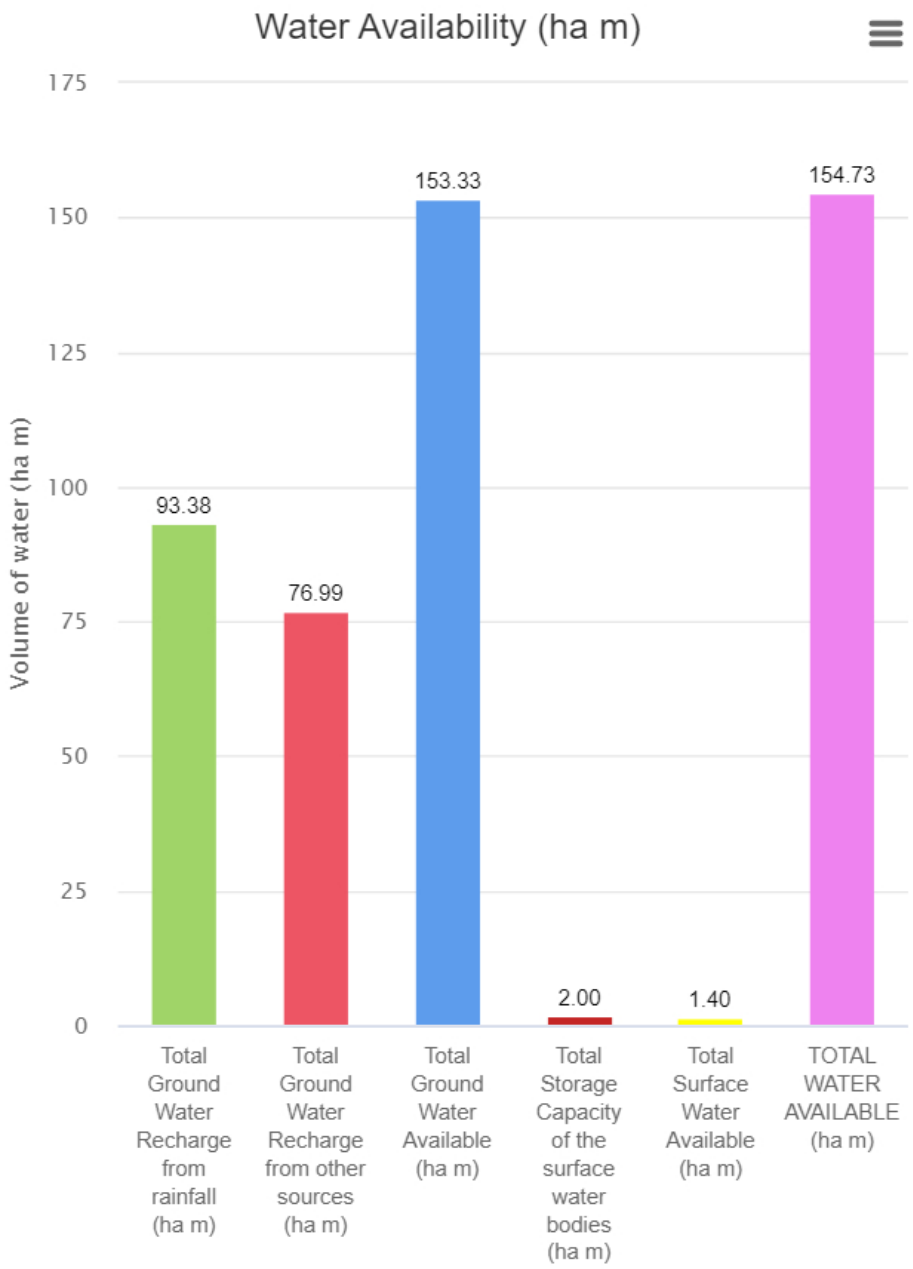


S.No.	Village Name	Location	Latitude	Longitude	Year	Well Type	Source	Well ID
1	MOTIBHAKHAR	Tarapsinh Hathi Singh Thakor	24.517919	72.32955	2022	Tube Well	ATAL JAL	BK_DAN_MOTIBHANKHAR_103
2	MOTI MAHUDI	Patel Danabhai Wajabhai	24.513902	72.32525	2022	Tube Well	ATAL JAL	BK_DAN_MOTI MAHUDI_24
3	MOTI MAHUDI	Dear Karshanbhai Bhikhabhai	24.515292	72.324849	2022	Tube Well	ATAL JAL	BK_DAN_MOTI MAHUDI_25
4	FATEGADH	Fardeen S Memon	24.52	72.3362	2022	Tube Well	ATAL JAL	BK_VADG_FATEGADH-216

S.No.	Well ID	Well Depth (mbgl)	Aquifer	PH	EC (µS/cm)	TDS (mg/L)	TH (mg/L)	Ca (mg/L)	Mg (mg/L)	Na (mg/L)	K (mg/L)	Carbonate (mg/L)	Bicarbonate (mg/L)	Sulphate (mg/L)	Chloride (mg/L)	Fluoride (mg/L)	Nitrate (mg/L)
1	BK_DAN_MOTIBHANKHAR_103	70.121951	UNCONFINED	8	1390	899.7	256	44.8	35	199	3.5	0	236	97	279.9	1.24	3.21
2	BK_DAN_MOTI MAHUDI_24	152.439024	CONFINED	7.8	2740	1705.5	252	64	22.4	512.2	2.6	0	230	140	729.7	1.32	3.25
3	BK_DAN_MOTI MAHUDI_25	106.707317	CONFINED	7.9	1570	1012.1	292	56	36.9	225	1.5	0	220	169	299.9	1.25	2.51
4	BK_VADG_FATEGADH-216	243.902439	CONFINED	7.5	980	590	186.1	30	27	128	5.2	0	183	0	224	2.16	22.3

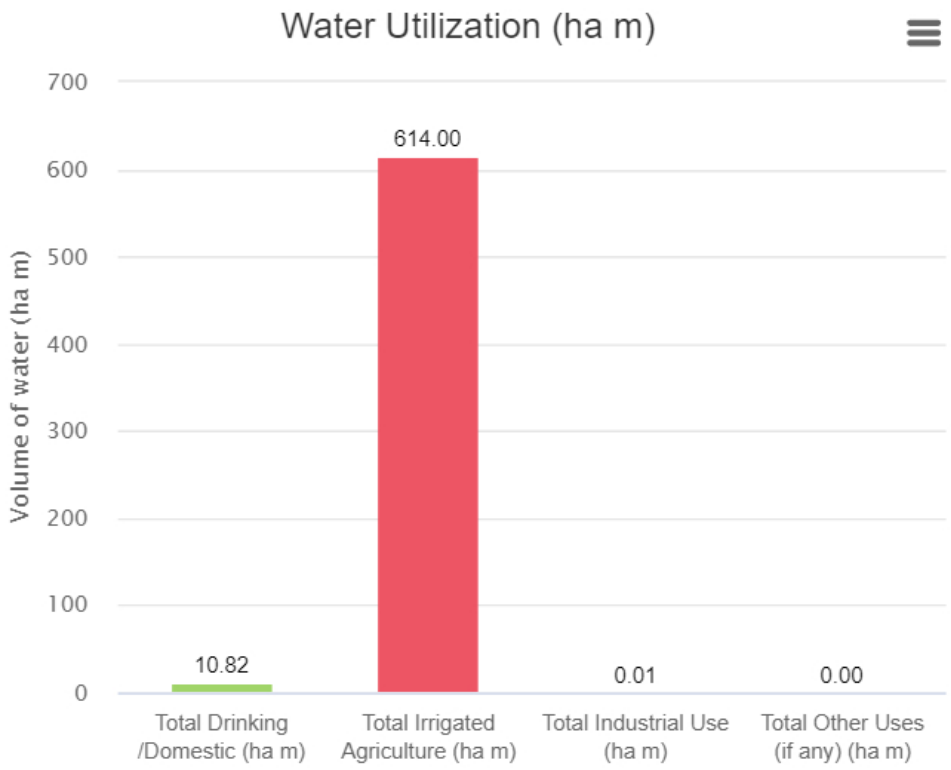
Water Availability Report

Water Availability		
1.	Total Available Water From Rainfall	
a.	Annual Rainfall (mm)	888.70
b.	Estimated Volume of Rainfall (ha m)	929.38
c.	Evapotranspiration rate (mm)	415.01
d.	Estimated Evapotranspiration losses (ha m)	434.01
e.	Estimated Water Available From Rainfall (ha m)	495.37
f.	Weighted Average Runoff Coefficient	0.10
g.	Rainfall Runoff(ha m)	49.54
2.	Ground Water	
a.	Recharge From Rainfall (ha m)	
(i)	Monsoon	93.38
(ii)	Non-Monsoon	0.00
	Total Ground Water Recharge from rainfall (ha m)	93.38
b.	Ground Water Recharge from other sources (ha m)	
(i)	Monsoon	35.49
(ii)	Non-Monsoon	41.50
	Total Ground Water Recharge from other sources (ha m)	76.99
c.	Losses Due to Natural Discharge (ha m)	17.04
	Total Ground Water Available (ha m)	153.33
d.	Ground Water Withdrawal (ha m)	183.12
3.	Surface Water	
a.	Surface Water available in storages	
(i)	Number of Surface Water Bodies and Storage Capacity (ha m)	
Type of Structures	No. of Structures	Storage Capacity (ha m)
Ponds	1	2
	Total Storage Capacity of the surface water bodies (ha m)	2.00
(ii)	Losses if Any (%)	30.00
(iii)	Surface Water Available in Storages (ha m)	1.40
b.	Transported Water from outside GP area (ha m)	0.00
	Total Surface Water Available (ha m)	1.40
4.	TOTAL WATER AVAILABLE	
	TOTAL WATER AVAILABLE (ha m)	154.73



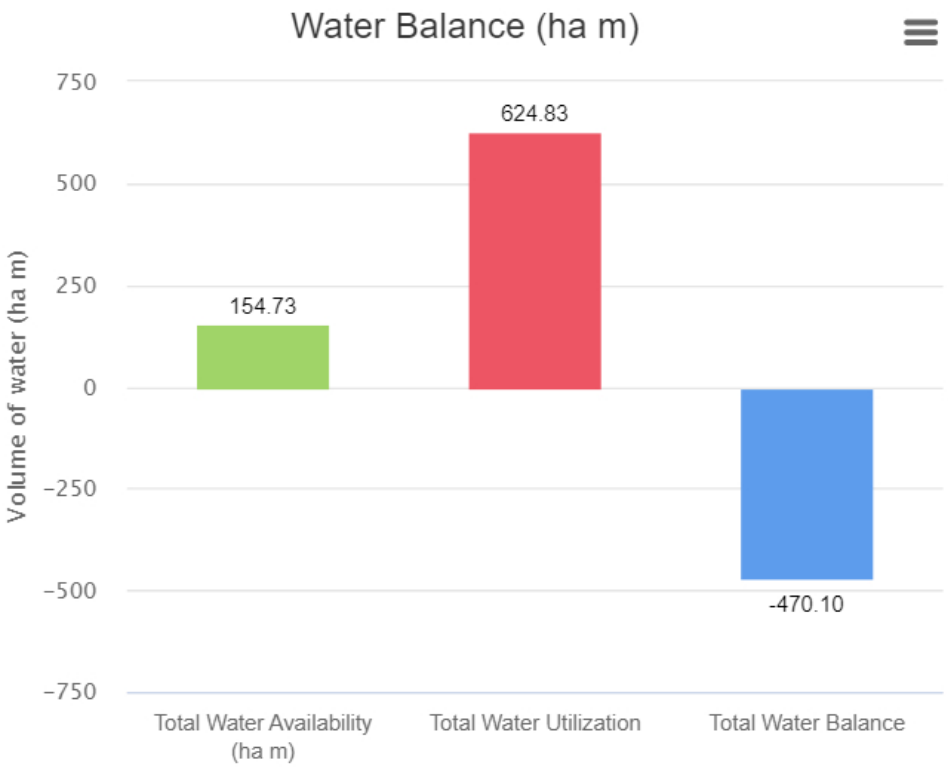
Water Utilization Report

Water Utilization		
A	Drinking /Domestic (ha m)	
i)	From Surface Water	0.13
ii)	From Ground Water	10.69
	Total Drinking /Domestic (ha m)	10.82
B	Irrigated Agriculture (ha m)	
i)	From Surface Water	0.00
ii)	From Ground Water	614.00
	Total Irrigated Agriculture (ha m)	614.00
C	Industrial Use (ha m)	
i)	From Surface Water	0.00
ii)	From Ground Water	0.01
	Total Industrial Use (ha m)	0.01
D	Other Uses (if any) (ha m)	
i)	From Surface Water	0.0000
ii)	From Ground Water	0.0000
	Total Other Uses (if any) (ha m)	0.0000



Water Balance Report

Water Balance		
A	Water Availability (ha m)	
i)	Total Ground Water Available	153.33
ii)	Total Surface Water Available	1.40
	Total Water Availability (ha m)	154.73
B	Water Utilization (ha m)	
i)	Total Surface Water Utilization	0.13
ii)	Total Ground Water Utilization	624.7
	Total Water Utilization	624.83
C	Balance [Surplus (+) / Deficit (-)]	
i)	Total Ground Water Balance	-471.37
ii)	Total Surface Water Balance	1.27
iii)	Total Water Balance	-470.10



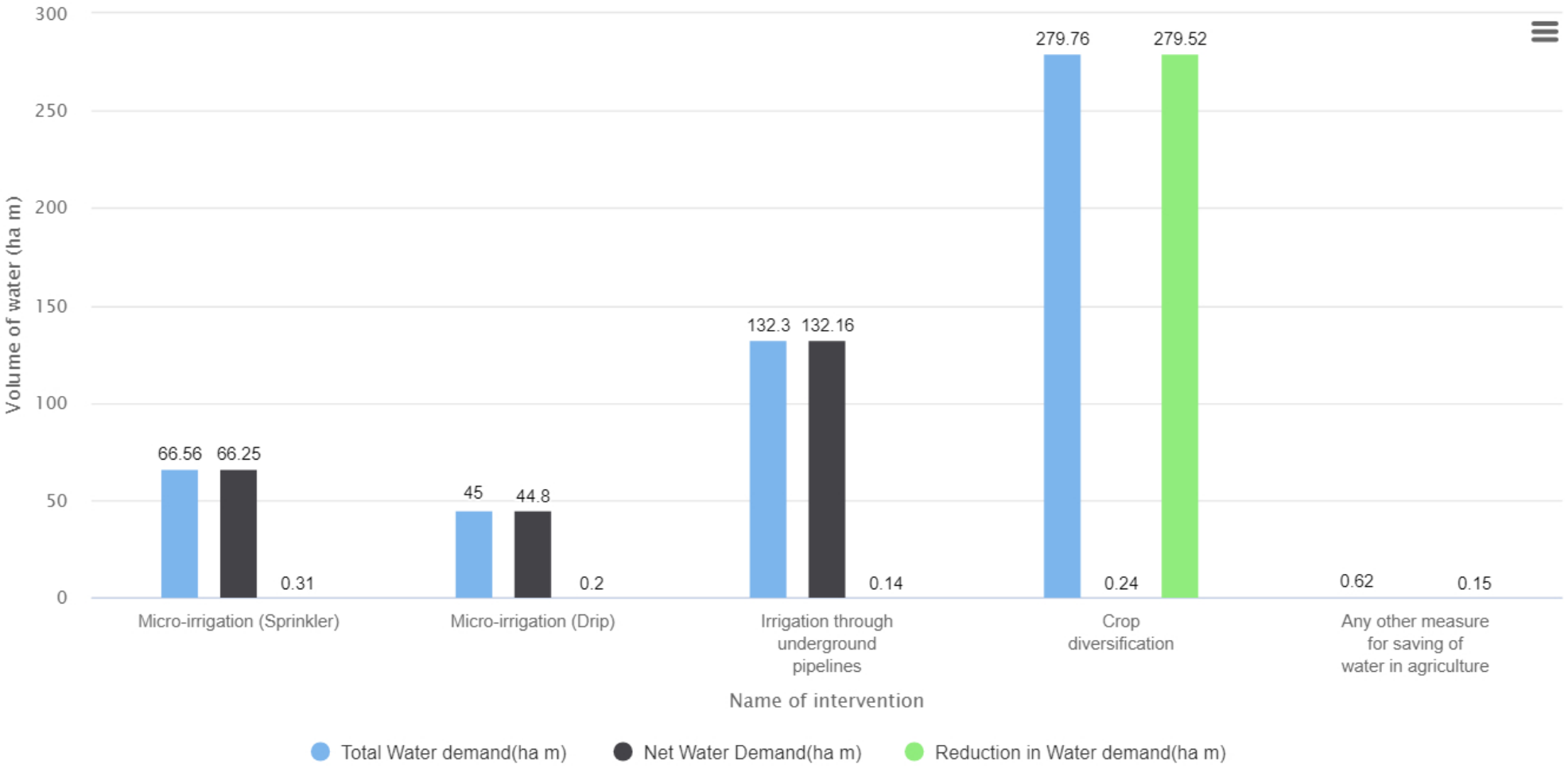
Water Budget Report

A	Allocation for Drinking & Domestic Requirements
I	Humans
i)	Present Human Population
ii)	Annual population growth rate (r)
iii)	No. of years
iv)	Projected human population (as in 2050)
v)	Per capita requirement (L)
vi)	Projected water requirement for humans (as in 2050) (ha m)
II	LiveStock
i)	Present livestock population
ii)	Annual population growth rate (r)
iii)	No. of years
iv)	Projected livestock population (as in 2050)
v)	Average per capita requirement (L)
vi)	Projected water requirement for livestock (as in 2050) (ha m)
B	Projected water requirement for humans and livestock (ha m) (as in 2050)
	Total Projected water requirement for humans and livestock (as in 2050) (ha m)
i)	Proposed share to be met from Ground water (in %)
ii)	Proposed share to be met from Surface water (in %)
iii)	Demand met from Ground water (ha m)
iv)	Demand met from Surface water (ha m)
C	Water available for future irrigation
i)	Ground Water
ii)	Surface water
iii)	Total water available for future irrigation

Demand Side

Demand Side						
Sr No	Name of Intervention	Proposed Area(ha)	Total Water demand(ha m)	Net Water Demand(ha m)	Reduction in Water demand(ha m)	Approximate cost(in lakh)
1	Sprinkler	2.00	66.56	66.25	0.31	2.10
2	Drip	1.00	45.00	44.80	0.20	1.20
3	Pipelines	2.00	132.30	132.16	0.14	0.30
4	Crop Diversification	1	279.76	0.24	279.52	0.06
5	Other Measures	1.00	0.62	--	0.15	0.21
	Total	7.00	524.24	243.45	280.32	3.87

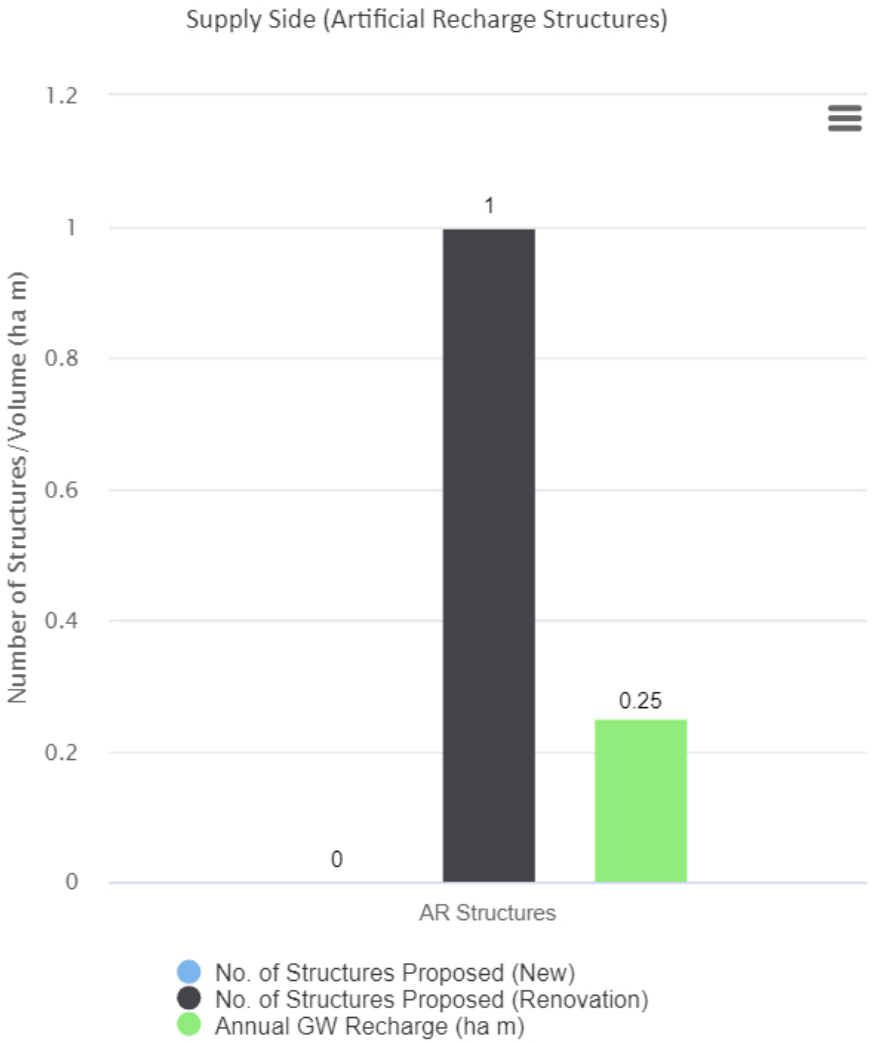
Funding Source						
Sr No	Name of Intervention	Area Proposed (ha)	Financial Year	Department Name	Scheme Name	Approximate Cost(in Lakh)
1	Sprinkler	2	2023-24	Gujarat Green Revolution Company Ltd. (GGRC)	Departmental Scheme	2.10
2	Drip	1	2023-24	Gujarat Green Revolution Company Ltd. (GGRC)	Departmental Scheme	1.20
3	UG Pipelines	2	2023-24	Agriculture & Farmer Welfare Department	Departmental Scheme	0.30
4	Crop Diversification	1	2023-24	Agriculture & Farmer Welfare Department	Departmental Scheme	0.06
5	Other Measures	1	2023-24	Agriculture & Farmer Welfare Department	Departmental Scheme	0.21
	Total	7				3.87

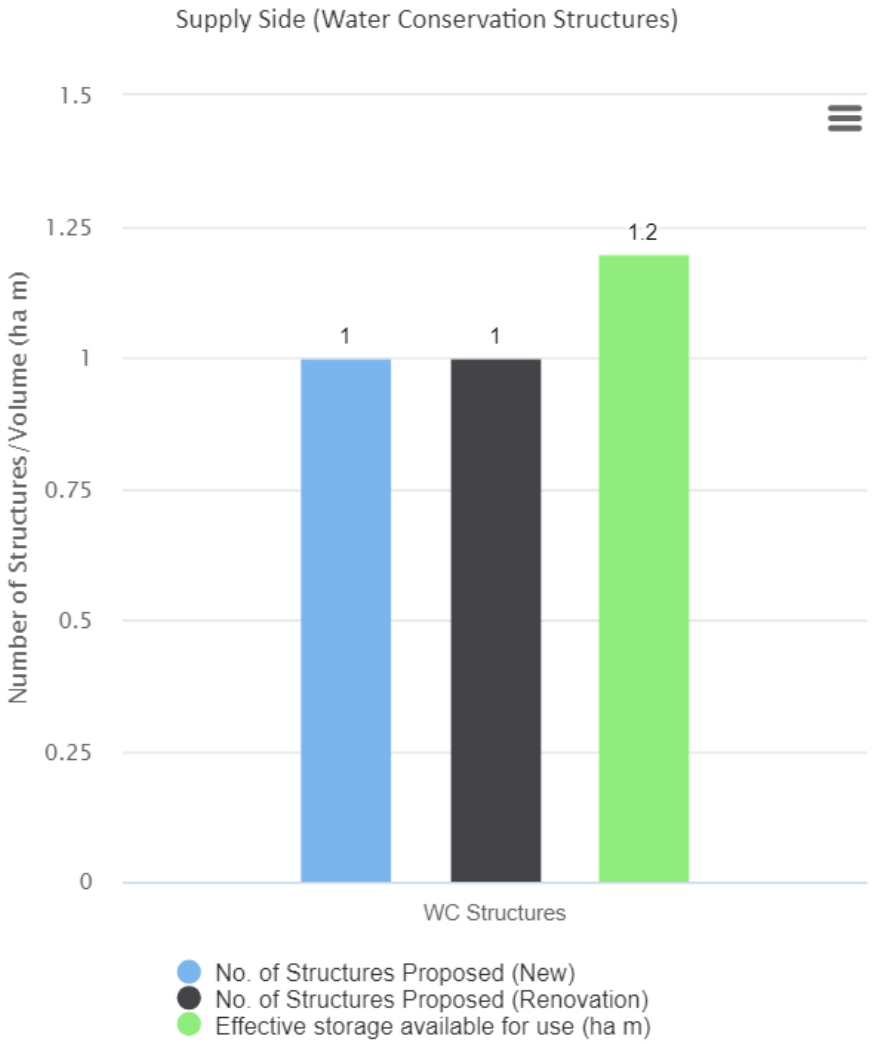

300
250
200
150
100
50
0
Volume of water (ha m)
66.56
66.25
0.31
45
44.8
0.2
132.3
132.16
0.14
279.76
0.24
279.52
0.62
0.15
Micro-irrigation (Sprinkler)
Micro-irrigation (Drip)
Irrigation through underground pipelines
Crop diversification
Any other measure for saving of water in agriculture
Name of intervention
Total Water demand(ha m)
Net Water Demand(ha m)
Reduction in Water demand(ha m)

Supply Side

Supply Side (Artificial Recharge Structures)										
SR No	Name of Structure	Storage Capacity(ha m)	Annual GW Recharge(ha m)	Village Name	Latitude	Longitude	Financial Year	Department Name	Scheme Name	Approximate Cost (in Lakhs)
1	DISILTING OF CHECKDAM 01 & 02(Renovation)	1	0.2500	Arkhi	24.524611	72.325274	2023-24	Gujarat Water Resources Development Corporation Ltd. (GWRDC)	Departmental Scheme	1.97
	Total	1.00	0.25							1.97

Supply Side (Water Conservation Structures)										
SR No	Name of Structure	Storage Capacity(ha m)	Effective storage available for use (ha m)	Village Name	Latitude	Longitude	Financial Year	Department Name	Scheme Name	Approximate Cost (in Lakhs)
1	Pond/Amrit Sarovar(Renovation)	1	0.6000	Arkhi	24.518966	72.328824	2023-24	Gujarat Water Resources Development Corporation Ltd. (GWRDC)	Departmental Scheme	1.06
2	Farm Pond(New)	1	0.6000	Arkhi	24.51780	72.31516	2023-24	Gujarat Water Resources Development Corporation Ltd. (GWRDC)	Departmental Scheme	2.00
	Total	2.00	1.20							3.06


Supply Side (Artificial Recharge Structures)
1.2
1
0.8
0.6
0.4
0.2
0
Number of Structures/Volume (ha m)
1
0.25
0
AR Structures
No. of Structures Proposed (New)
No. of Structures Proposed (Renovation)
Annual GW Recharge (ha m)


Supply Side (Water Conservation Structures)
1.5
1.25
1
0.75
0.5
0.25
0
Number of Structures/Volume (ha m)
1
1
1.2
WC Structures
No. of Structures Proposed (New)
No. of Structures Proposed (Renovation)
Effective storage available for use (ha m)

PGWM Committee

PGWM Committee (VWSC/WUA/CBO)				
Sr.no	Name	Designation	Gender	Mobile /Landline No.
1	VADHDA AASHABEN MOTIBHAI	MEMBER	Female	814151525
2	ODHAVIYA NAGABHAI JAGMALBHAI	MEMBER	Male	8141515254
3	GARASIYA CHUNABHAI	MEMBER	Male	8141515254
4	DEVDA DASHARTHKUMAR A	SARPANCH	Male	8141515254
5	PATEL BABUBHAI	MEMBER	Male	8690078821
6	DEVDA MADANSINH ARJUNSINH	MEMBER	Male	814151525
7	CHAUDHREE R F	TCM	Male	8141515254

Details of PGWM Meeting in which resolution was passed												
Sr.no	State	District	Block	Gram Panchayat	Date	GP Name	Village Name	Meeting Location	Male	Female	Transgender	Total number of attendees
1	Gujarat_24	Banaskantha_441	Dantiwada_3735	Arkhi_155944	null	Arkhi_155944	Arkhi	PANCHAYAT	10	5	0	15

Photograph